PARISHRAM 2.0 2026 (Class 12th)

CHEMISTRY

DPP 02

d & f-Block Elements

Q1 The transition element having least enthalpy of atomisation is

(A) $\mathbf{Zn}$

(B) $\mathbf{V}$

(C) $\mathbf{Fe}$

(D) $\mathbf{Cu}$

Q2 The third ionization enthalpy is minimum for:

(A) $\mathbf{Ni}$

(B) $\mathbf{Co}$

(C) $\mathbf{Mn}$

(D) $\mathbf{Fe}$

Q3 The transition metals exhibit higher enthalpies of atomization due to:

(A) Their ability to show variable oxidation states

(B) The presence of incompletely filled dsubshell

(C) Their ability to exist in the solid state with unpaired electrons

(D) Strong interatomic interaction aries because of having large number of unpaired electrons in their atoms

Q4 Amongst the following ions, which is considered as most stable in $\mathbf{M^{2+}}$ state?

(A) $\mathbf{Ti^{2+}/Ti(-1.63\ V)}$

(B) $\mathbf{V^{2+}/V(-1.11\ V)}$

(C) $\mathbf{Cr^{2+}/Cr(-0.90\ V)}$

(D) $\mathbf{Mn^{2+}/Mn(1.18\ V)}$

Q5 The catalytic activity of transition metals and their compounds is mainly due to

(A) Their magnetic behavior

(B) Their unfilled d-orbitals

(C) Their ability to adopt variable oxidation state

(D) Their chemical reactivity

Q6 The ability of d-block elements to form complexes is due to

(A) Large and highly charged ions

(B) Vacant low energy orbitals to accept lone pair of electrons from ligands

(C) Low polarizing power of cation

(D) None is correct

Q7 Transition elements act as catalyst because:

(A) their melting points are low.

(B) their ionisation enthalpy values are high.

(C) they have low density.

(D) they can show variable oxidation states.

Q8 Complex forming tendency of an element depends upon

(A) High charge

(B) Small size of its cation

(C) Availability of vacant d-orbitals

(D) All of these

Answer Key

Q1	A
Q2	D
Q3	D
Q4	A
Q5	C
Q6	B
Q7	D
Q8	D

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

Zn

Video Solution:

Q2 **Text Solution:**

The third ionization enthalpy is minimum for Fe.

Video Solution:

Q3 **Text Solution:**

Many unpaired (n – 1)d electrons from neighbouring atoms overlap to form very strong metallic bonds, so great energy is needed to separate the atoms, giving transition metals their characteristically high enthalpies of atomisation.

Video Solution:

Q4 **Text Solution:**

$\mathrm{Ti}^{2+}/\mathrm{Ti}(-1.63\ \mathrm{V})$

Video Solution:

Q5 **Text Solution:**

Transition metals can exhibit multiple oxidation states. This flexibility allows them to bind and activate reactant molecules in different ways, lowering the activation energy of reactions. Their partially filled d orbitals also facilitate complex formation and redox processes, key factors in catalytic mechanisms. Thus, variable oxidation states is the principal reason for their catalytic activity.

Video Solution:

Q6 **Text Solution:**

The ability of **d-block elements** to form complexes is mainly due to the presence of **Vacant low energy orbitals to accept lone pair of electrons from ligands**

Video Solution:

Q7 **Text Solution:**

They can show variable oxidation states.

Video Solution:

Q8 **Text Solution:**

Complex forming tendency of an element depends upon high charge, small size of its

cation,

cant d-orbitals. **Video**

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